

# Fatima Al-Rashid

SENIOR DATA SCIENTIST · AI/ML RESEARCH LEAD ·  
NLP SPECIALIST

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Location: Chicago, IL 60601

LinkedIn: linkedin.com/in/fatima-  
alrashid-ds

Google Scholar:  
scholar.google/fatima-alrashid

GitHub: github.com/falrashid-ml

12+  
Yrs Experience

34  
Papers Published

5,800+  
Citations

h-index: 18  
Google Scholar

\$310M  
Value Delivered

6  
Patents Filed

## PROFESSIONAL SUMMARY

Pioneering AI researcher and senior data scientist with **12+ years** at the intersection of deep learning, natural language processing, and real-world product deployment. Proven track record publishing in top-tier venues (NeurIPS, ICML, ACL, EMNLP) while simultaneously delivering \$310M+ in measurable business value across healthcare, finance, and enterprise software. Expert in LLMs, transformer architectures, causal inference, and MLOps at scale. Led 4-person to 35-person data science teams at Google, IBM Research, and Tempus AI. Passionate about responsible AI development, fairness/bias mitigation, and bridging the gap between cutting-edge research and production systems. Fluent Arabic and English speaker; published work cited by OpenAI, DeepMind, and Meta AI.

## CONTACT

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## ML SKILLS

### LLMs / Transformers

GPT, BERT, T5, Llama, Mistral

### NLP / Text Mining

Parsing, NER, Summarization, QA

### Causal Inference

DoWhy, CausalML, IV, DiD

### Computer Vision

CNNs, ViT, Detection, Segm.

### Reinforcement Learning

RLHF, PPO, DQN, A3C

### Statistical Modeling

Bayesian, Mixed Effects, GLM

### Graph Neural Networks

PyG, DGL, GCN, GAT

### MLOps / LLMops

MLflow, W&B, BentoML, vLLM

## PROGRAMMING

### Python

### R / Julia

### SQL

### Scala (Apache Spark)

### C++ (Performance Tuning)

## PLATFORMS

PyTorch JAX

## WORK EXPERIENCE

### Principal Data Scientist / AI Research Lead

Feb 2021 - Present

Google LLC · Chicago, IL (Hybrid)

- Lead 35-person AI research team spanning LLM fine-tuning, multimodal understanding, and search ranking; \$42M research budget; partner with Google DeepMind on 3 joint research programs
- Architected Google Search's query intent classification system processing 8.5B daily queries; improved CTR by 14% through semantic understanding improvements — ~\$420M annual revenue impact
- Led "Project Chimera" — large-scale multilingual document understanding system covering 92 languages; deployed to Google Docs, Gmail, and Google Drive serving 3B+ users
- Designed responsible AI evaluation framework (bias, toxicity, factuality) adopted as mandatory review for all Google AI product launches; reduced harmful output incidents by 78% in first year
- Filed 4 patents on efficient transformer attention mechanisms; two now incorporated in production TPU v4 inference optimization
- Published 8 papers at NeurIPS (2021, 2022, 2023), ICML (2022), and ACL (2023) as Google Research author; combined 2,200+ citations in 3 years
- Mentored 12 research scientists to senior/staff level; built internal ML literacy program reaching 800+ Google engineers annually

PyTorch JAX TPU/XLA Flax BigQuery Vertex AI Python

### Senior Data Scientist - Clinical AI

Sep 2017 - Jan 2021

Tempus AI · Chicago, IL

- Built ML models for genomic data analysis and treatment recommendation across 3 oncology types; deployed at 430+ hospital systems treating 200K+ cancer patients annually
- Developed NLP pipeline for unstructured EHR data extraction (pathology reports, clinical notes); achieved 96.2% F1 on benchmark, surpassing human reviewer baseline of 91.8%
- Led FDA 510(k) submission for Tempus's AI-assisted tumor mutation burden test; first ML-based molecular test cleared for clinical use in 2020 at Tempus
- Designed causal inference framework to measure treatment effect heterogeneity across patient subpopulations; used by oncologists at Memorial Sloan Kettering and MD Anderson
- Built real-time clinical trial matching system using BERT fine-tuning; matched 18,000+ patients/year to relevant trials, improving enrollment rates by 34%

PyTorch HuggingFace Spark Python AWS Redshift

### Research Scientist

Jul 2013 - Aug 2017

IBM Research - Watson · Yorktown Heights, NY

## SPEAKING

### NeurIPS 2023

Invited Talk:  
"Responsible LLMs at Scale"

### ICML 2022

Keynote:  
"Causal ML in Healthcare"

### ACL 2021

Panel Chair:  
NLP for Healthcare

### Strata 2020

Keynote: "AI in Clinical Genomics"

### TEDxChicago

"AI as a Tool for Health Equity" · 140K views

## UPCOMING

ICLR 2025 · Area Chair  
ACL 2025 · PC Member  
NeurIPS 2025 · Reviewer

## MEDIA

**Wired** - "The AI Researchers Fixing Bias" (2022)

**MIT Tech Review** - 35 Innovators U35 Profile (2020)

**Nature News** - Quoted Expert: Clinical AI (2023)

**Forbes** - "AI Women to Watch" (2021)

**Podcast:** TWIML AI Podcast Ep. 482 (2021)

**Podcast:** Lex Fridman Podcast Ep. 340 (2022)

## ACADEMIC SERVICE

Adj. Professor, UChicago CS  
Area Chair: NeurIPS, ICML

TensorFlow

HuggingFaceLangChain

LlamaIndexRaySpark

DatabricksdbtAirflow

PrefectAWS SageMaker

GCP Vertex AISnowflake

BigQueryMLflowW&B

CERTIFICATIONS

- ★ AWS ML Specialty  
AWS · 2022
- ★ GCP Professional ML Engineer  
Google Cloud · 2022
- ★ DeepLearning.AI Specialization  
Coursera · 2020
- ★ Databricks ML Associate  
Databricks · 2021
- ★ NVIDIA Deep Learning Cert  
NVIDIA DLI · 2021
- ★ Stanford ML Certificate  
Stanford Online · 2019

EDUCATION

**Ph.D., Computer Science**  
**University of Illinois Urbana-Champaign**  
Champaign, IL · 2013  
Dissertation: "Hierarchical Bayesian Models for Structured Prediction in NLP" · NSF Graduate Fellow · Advisor: Prof. Julia Hockenmaier

**M.S., Statistics**  
**American University of Beirut**  
Beirut, Lebanon · 2009  
Concentration: Machine Learning · GPA: 4.0/4.0

**B.S., Mathematics & Comp Sci**  
**King Abdullah University (KAUST)**  
Thuwal, Saudi Arabia · 2007  
Valedictorian · Full Scholarship

LANGUAGES

|         |                   |
|---------|-------------------|
| Arabic  | Native (C2)       |
| English | Native (C2)       |
| French  | Fluent (C1)       |
| German  | Intermediate (B1) |

AWARDS

- 🏆 NeurIPS Outstanding Paper (2021)
- 🏆 ACL Best Paper Award (2019)
- 🏆 MIT Tech Review 35U35 (2020)
- 🏆 Forbes 30U30 – Enterprise Tech (2019)
- 🏆 NSF CAREER Award (\$500K) (2018)
- 🏆 DARPA AI Researcher Award (2017)

- Core researcher on IBM Watson's question-answering engine; contributed 6 major algorithm improvements that boosted Jeopardy!-domain accuracy from 72% to 89%
- Led Knowledge Graph construction team building Watson's domain ontologies; developed entity resolution algorithms with 94.7% precision across 200M+ entity pairs
- Published 14 papers at top venues (ACL, EMNLP, NAACL, AAAI); 3 received Best Paper nominations; h-index grew from 3 to 12 during tenure
- Received NSF CAREER Award (\$500K, 2018) and DARPA Emerging Researchers in AI Award (2017) based on IBM research contributions

Python

TensorFlow

NLTK

spaCy

Java

Scala

Research Intern → Junior Scientist  
Microsoft Research (MSR) · Redmond, WA

Jun 2011 – Jun 2013

- Intern (2011) then Research Intern (2012) working on MSR's NLP group; co-authored 4 papers on semantic role labeling and coreference resolution during PhD
- Developed Arabic NLP toolkit (open-sourced; 800+ GitHub stars) covering tokenization, POS tagging, and named-entity recognition for Modern Standard and dialectal Arabic

RESEARCH PROJECTS

|                                               |                                                                                                                                                                                                     |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MedLLM</b><br>2022–2023 · Google           | Fine-tuned Flan-T5 for clinical question answering on 2M+ de-identified EHR records. Achieved SOTA on MedQA-USMLE (86.4%) and PubMedQA (92.1%). Featured in <i>Nature Medicine</i> (2023).          |
| <b>CausalBench</b><br>2020–2021 · Tempus      | Built open benchmark for evaluating causal inference methods in healthcare settings. 12 datasets, 8 evaluation metrics. Adopted by Stanford CRUMB lab and 40+ research groups. GitHub: 1,900 stars. |
| <b>ArabiNLP Toolkit</b><br>2012–Present · OSS | Comprehensive Arabic NLP library covering 8 dialects + MSA. 800+ GitHub stars, used in 60+ academic papers. Integrated into HuggingFace Datasets hub as Arabic benchmark reference.                 |

SELECT PUBLICATIONS (34 TOTAL · 5,800+ CITATIONS · H-INDEX: 18)

- NeurIPS 2023** "Efficient Sparse Attention for Long-Context LLMs" · Al-Rashid, Chen, et al. · 380 citations
- Nature Medicine 2023**"Clinical QA with Domain-Adapted Language Models" · Al-Rashid et al. · 210 citations
- ICML 2022** "Scalable Causal Discovery for Healthcare Observational Data" · Al-Rashid, Park, et al. · 490 citations
- NeurIPS 2021** "Towards Unbiased Arabic NLP" – **Outstanding Paper Award** · 640 citations

**ACL 2019** "Multi-lingual Coreference Resolution" – **Best Paper Award** · 820 citations

**EMNLP 2016** "Cross-Lingual Transfer for NER" · Al-Rashid, Liu · 710 citations

PARSER EXTRACTION TEST TABLE

| Field              | Raw Value                                   | Parsed Format         | Status    |
|--------------------|---------------------------------------------|-----------------------|-----------|
| name               | Fatima Al-Rashid                            | First Compound-Last   | ✓<br>HIGH |
| email              | fatima.alrashid@airesearch.io               | RFC 5321              | ✓<br>HIGH |
| phone              | +1 (312) 944-0827                           | E.164 / NANP          | ✓<br>HIGH |
| address            | 820 N Michigan Ave, Chicago, IL 60601       | Street City State ZIP | ✓<br>HIGH |
| current_title      | Principal Data Scientist / AI Research Lead | Multi-value title     | ✓<br>HIGH |
| employer           | Google LLC                                  | Legal entity name     | ✓<br>HIGH |
| work_history_count | 4 positions, 5 employers                    | Array[Experience]     | ✓<br>HIGH |
|                    | Mon YYYY – Mon YYYY /                       |                       | ✓         |

PC Member: ACL, EMNLP, NAACL  
Reviewer: Nature MI, JMLR, TACL  
Ph.D. Committee: 8 students  
NSF Grant Panelist (2019–Present)

VOLUNTEERING

Women in ML Org (WiML) – Board Member (2019–Present)  
Black in AI – Mentor (2018–Present)  
Girls Who Code – AI Curriculum  
UN AI for Good Advisor (2022)

PATENTS

- US-2022-0318492**  
Sparse attention for long documents
- US-2022-0301847**  
Multi-lingual embedding alignment
- US-2021-0244123**  
Causal graph discovery from EHR
- US-2021-0207441**  
Clinical trial matching NLP system
- US-2020-0176834**  
Entity resolution across biomedical ontologies

INTERESTS

Arabic calligraphy & typography  
Marathon running (4 completed)  
Vegetarian cooking instructor  
Chess (FIDE Elo: 1840)  
Arabic children's book author

|                   |                                                                            |                       |           |
|-------------------|----------------------------------------------------------------------------|-----------------------|-----------|
| date_range_format | Present                                                                    | MMM YYYY              | HIGH      |
| total_years_exp   | 12 (2013–2025)                                                             | Integer               | ✓<br>HIGH |
| highest_degree    | Ph.D., Computer Science                                                    | Degree, Field         | ✓<br>HIGH |
| education_count   | 3 institutions                                                             | Array[Education]      | ✓<br>HIGH |
| skills_ml         | LLMs, NLP, Causal Inference, CV, RL, GNN, MLOps                            | Array[Skill]          | ✓<br>HIGH |
| languages         | Arabic(Native), English(Native), French(Fluent), German(B1)                | Lang:Level pairs      | ✓<br>HIGH |
| publications      | 34 papers, 5800+ citations, h-index 18                                     | Publications metadata | ✓<br>HIGH |
| patents           | 6 patents filed (4 at Google, 2 co-inventor)                               | IP entries            | ✓<br>HIGH |
| awards            | NeurIPS Outstanding (2021), ACL Best Paper (2019), MIT 35U35, Forbes 30U30 | Award[name,year]      | ✓<br>HIGH |